

Modeling of High Risk Errors of Certification Error in the NSLP

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Final Briefing to USDA Food and Nutrition Service
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Study Objectives

- 1. Develop a statistical model to identify LEAs at high risk for certification error**
- 2. Develop a monitoring tool for States to use for selecting SFAs for Additional Administrative Reviews (AARs)**
 - SFAs at high risk for certification error
 - SFAs that meet other FNS selection criteria

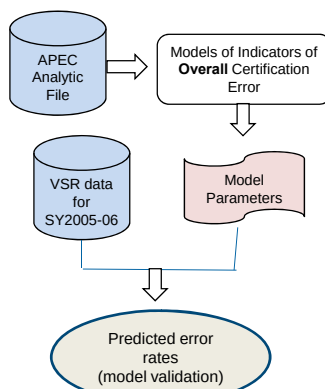
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Overview of Project

Phase 1 Create Analysis Files

- **APEC** –new variable construction (already linked with extant data)
- **RORA** – link with extant data (CCD, LAUS, SAIPe) & construct vars
- **VSR** – link with extant data (CCD, LAUS, SAIPe) & construct vars

Phase 2 Model High-Risk Indicators of Certification Error



* RORA data were also used for modeling.

Phase 3 Develop Web-based Risk Assessment Tool

Replicate SAS/STATA analysis work in SQL query language.

Develop administrator functions for loading data files.

Develop reports for State Child Nutrition agencies.

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Challenges for the Future

- **Creating analysis files**
 - SFAIDs in VSR are not consistent over time and do not match Department of Education CCD files
 - Some features less effective if link file is not maintained
- **Model development**
 - Model is based on APEC data (single year)
 - Results may become less applicable over time
- **Tool design**
 - Some AAR selection criteria are not measured well in FNS annual data collections
 - Future SFA guidance and data collection plans should be developed hand-in-hand with tool revisions/updates

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Creating Analysis Files

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Sources of Certification Error Measures

Data Source	Time Period	Certification Error Measure		
		Overall	Household	Administrative
Access, Participation, Eligibility and Certification Study (APEC)	SY2005-06	√	√	√
Regional Office Review of Applications (RORA)	Annual since SY2004-05			√
Verification Summary Reports* (VSR)	Annual since SY2004-05	√		

* VSR measures of certification error were not used for statistical modeling because verification samples are not random.

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Sources of Explanatory Variables

Data Source	Types of Variables
FNS, Verification Summary Reports (VSR)	SFA application, eligibility, and verification information
Department of Education, Common Core of Data (CCD) ^a	SFA characteristics (grades, student-teacher ratios, administrator-student ratios, finances)
Census Bureau, Small Area Income and Poverty Estimates (SAIPE)	Poverty estimates for school districts and counties
Bureau of Labor Statistics, Local Area Unemployment Statistics (LAUS)	County unemployment rates
FNS, Regional Office Review of Applications (RORA)	NSLP features
FNS National Datafile	State NSLP meal counts

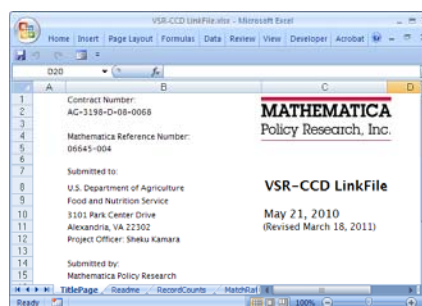
^a CCD data were taken from the Local Education Agency Universe Survey, Local Education Agency Finance Survey, and Public Elementary/Secondary School Universe Survey.

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Creating Analysis Files

- **Challenges:**
 - SFAIDs ≠ LEAIDs
 - SFAIDs are not consistent over time
- **Solution = VSR-CCD Linkfile**
 - 96% of SFAs were matched with CCD via deterministic and probabilistic matching
- **Ongoing Challenge**
 - Maintaining the link file using the functions in the monitoring tool
 - Linking is needed for mapping; link file may be useful for other research projects



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Methods for Developing Statistical Models

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Selecting the Model Specification

- Estimated with Tobit regression models

$$(1) \text{Error}_{kj} = \max \left[0, X_{kj} \beta_{kj} + u_{kj} \right]$$

- Main criteria for model selection was predictive power
 - Apply model to data other than those used to estimate the model
- Also examined goodness of fit

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Considered Alternative Model Specifications

- **Certification Error Measures**
 - Error rates vs. levels (dollar amounts)
 - Aggregate vs. disaggregate measures
- **Selecting Explanatory Variables**
 - High policy-relevance
 - Strong observed correlation with certification error
- **Model estimation technique**
 - OLS
 - Separate models based on enrollment category

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Certification Error Measures

- **Aggregate measures are based on all reimbursements**
- **Disaggregate measures are based on reimbursements to**
 - Certified NSLP-free but not F/RP eligible (FCNE)
 - Certified NSLP-free but eligible for NSLP-RP (FCRE)
 - Certified NSLP-RP but not F/RP eligible (RCNE)
 - Certified NSLP-RP but eligible for NSLP-free (RCFE)
 - Not certified but eligible for NSLP-free (NCFE)
 - Not certified but eligible for NSLP-RP (NCRE)

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Selecting Explanatory Variables

Two-Pronged approach:

1. Select a set of **core variables** based on policy relevance or theoretical relationship to certification error
2. Use an **automated process** to add additional variables based on their observed correlation with certification error

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Core Explanatory Variables

Type of Variable	Variables
Verification Variables	SFA uses random verification sample Percent of applications with benefits changed due to verification Verification nonresponse Interactions with random sample
Certification Variables	Percent of certified students: • Certified categorically • Certified without application
SFA Characteristics	Enrollment Percent of all students: • Certified for NSLP-free • Certified for NSLP-RP

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Automated Process for Variable Selection

- Include core variables and add variables with the highest correlation to certification error measure
- Continue adding until the next variable is not significant
 - Other procedures were considered
- Typically 2 or 3 variables were added
 - Most explanatory variables come from the core set

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Assessment of Model Performance

- 1.** Estimate model using APEC or RORA data
- 2.** Apply model to VSR data to get predicted error for all districts
- 3.** Combine district predictions to get national estimate
- 4.** Compare imputed national VSR estimates to national estimates from APEC and RORA

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Results of Statistical Modeling

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Final Model Specifications

- All models are based on rates of error (not levels)
 - Predicted error rates are translated into predicted error levels using meal reimbursement estimates
- All models draw covariates solely from VSR
- Models of overall and household reporting error from APEC are based on disaggregated measures
- Models of administrative error from APEC and RORA are based on aggregate measures

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Model Performance is Strong

Distribution of Actual and Imputed Certification Error Rates

Measure	Type	Mean	25 th	75 th
APEC Overall Certification Error Rate	Actual	9.4	5.6	11.8
	Imputed	8.9	4.8	12.1
APEC HH Reporting Certification Error Rate	Actual	7.7	4.5	10.3
	Imputed	7.2	3.9	9.8
APEC Admin. Certification Error Rate	Actual	1.5	0.0	1.4
	Imputed	1.1	0.0	1.8
RORA Admin. Certification Error Rate	Actual	1.3	0.2	1.9
	Imputed	1.5	0.8	2.1

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SFA Characteristics Associated with Higher Error Rates

- **Overall and Household Reporting Error**
 - Larger districts
 - Districts not using random verification samples and with higher levels of benefit reduction through verification
 - Districts not using random verification samples and with higher levels of verification nonresponse
 - Districts with lower levels of students certified through categorical eligibility
- **APEC Administrative Error**
 - Verification results
- **RORA Administrative Error**
 - Enrollment

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Using Statistical Models in the Monitoring Tool

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Purpose of the Monitoring Tool

Identify SFAs Meeting Criteria for AARs

1. LEAs with high risk of certification error

- Based on statistical models

2. Other criteria developed by FNS

- High level of verification non-response or termination
- High claiming rates
- Establishment of a Provision 2/3 base year

- Need definition of “high risk” LEAs from the statistical models

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Desirable Characteristics of Risk Definition

- **Simple**

- Straightforward to incorporate into the monitoring tool and easy to understand by tool users

- **Targeted**

- Identifies a set of districts accounting for a disproportionate share of national erroneous payments due to certification error

- **Responsive to time-variant characteristics**

- Does not identify the same set of districts as high risk in every year

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Step 1: Translate Predicted Error Into Risk Categories

- **Threshold for high risk**

- $\$50,000 * (\text{State Median Enrollment} / \text{National Median})$
- Corresponds to 85th percentile nationally
- State adjustment accounts for across State differences in district size

- **Threshold for medium risk**

- $\$25,000 * (\text{State Median Enrollment} / \text{National Median})$
- Corresponds to 65th percentile nationally

- **Error Rate Floor**

- Avoid categorizing large districts with low error rate as high risk
- Presentation uses a 5% floor

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Step 2: Translate Error Rates Into Risk Scores

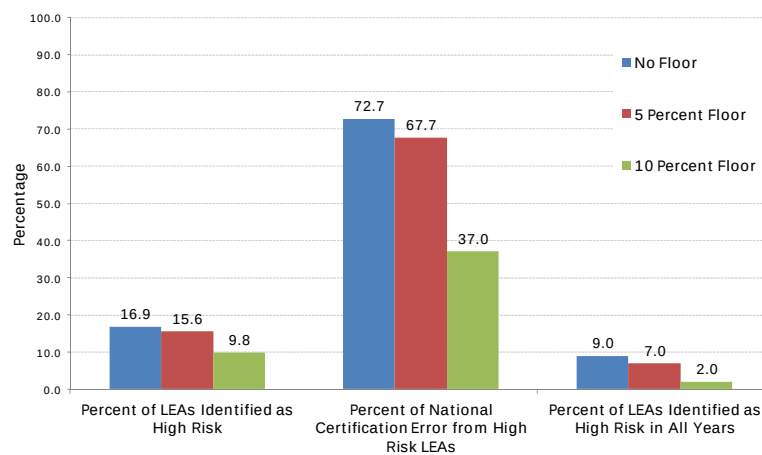
$$\text{Risk Score} = \Phi \left(\frac{\text{Estimated Dollar Amount In Error} - \text{State Threshold for High Risk}}{\text{Standard Deviation of Estimated State Dollar Amount In Error}} \right) * 100$$

- Districts' scores are on a scale of 0 to 100
- Risk scores allow comparisons over time and across states

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Prevalence and Targeting of Risk Definitions with Different Rate Floors



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Characteristics of Districts Characterized as High Risk

- **Less favorable verification results**
 - Higher error and nonresponse rates
- **Less certification through categorical eligibility and more not subject to verification**
- **Much larger enrollment**
 - About 12,000 for high risk compared to about 1,000 for low risk
 - Related to focus on levels rather than rates
 - Importance of rate floor

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Design of the Monitoring Tool

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Key Functions

1. FNS: Load data annually and generate reports
2. States: View reports and select SFAs for AARs

<http://nslp.relyonmedia.com/>

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Conclusions

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Study Achievements

- **Models are predictive of certification error**
- **Models summarize relationship between district characteristics and certification error**
 - Importance of verification results and enrollment
- **Models are informative for assessing risk in future years**
 - Simple, targeted, and time-responsive risk definition
- **The tool synthesizes a substantial amount of information in a user-friendly way**

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Study Limitations

- **Results may become less applicable over time**
 - Requires future studies of certification error such as APEC
- **Statistical models do not perfectly predict certification error**
- **Tool relies on imputed values for meal counts**
 - Allows States to replace imputations with actual counts

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